WORKED SOLUTION

STEP III 1999, Question 9

Mechanics • Gravitational force

STEP III 1999, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q9

This question has a full guided treatment in *Beyond STEP*, Section 10.7, *Gravity Inside a Shell*: https://beyondstep.keithmcnulty.org/ch10/shell-theorem.html

The key idea

Gravity from an extended body is the vector sum of the pulls of its small pieces. For the ring, symmetry does most of the work: pieces on opposite sides of the ring cancel each other's sideways pulls, and every piece is the same distance from the particle, so only one simple component survives. For the shell, we slice it into rings whose axis passes through the particle, use the ring result for each slice, and add the slices up with an integral. A well-chosen substitution makes that integral collapse to zero.

Solution

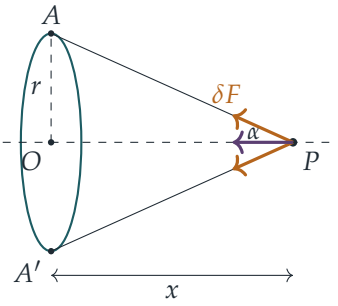
The ring

Model the ring as a thin uniform circle of mass m and the particle as a point of unit mass at P, on the axis at distance x from the centre O. Every point of the ring is at the same distance

$$d = \sqrt{x^2 + r^2}$$

from P, by Pythagoras in the right-angled triangle formed by P, O and that point.

Direction. Pair each small element of the ring at A with the equal element at the diametrically opposite point A'. They are the same distance d from P, so they pull with equal magnitudes along PA and PA'. These two lines are reflections of each other in the axis, so the components perpendicular to the axis cancel, and the components along the axis add, both pointing from P towards O. The whole ring is made of such pairs, so the net force on the particle is along the axis, towards the centre of the ring.



Magnitude. An element of mass δm pulls with force $G\,\delta m/d^2$ along PA. If α is the angle between PA and the axis PO, then $\cos\alpha = x/d$, so its component towards O is

$$\frac{G\,\delta m}{d^2}\cdot\frac{x}{d} = \frac{Gx\,\delta m}{(x^2+r^2)^{3/2}}.$$

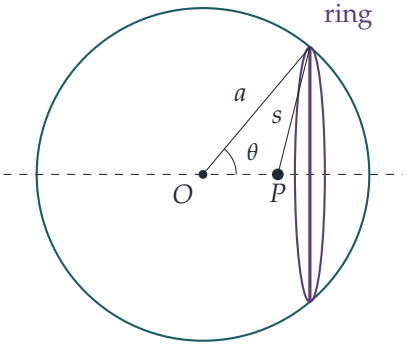
The factor multiplying δm is the same for every element, so summing over the ring just replaces δm by m:

$$F = \frac{Gmx}{(x^2+r^2)^{3/2}},$$

directed towards the centre, as required. If P is on the other side of the ring this holds with x replaced by $|x|$; in the next part we use x as a signed distance, and then the formula with signed x gives the signed component of the force correctly.

The shell

Let the shell have centre O, radius a and mass M, so its surface density is $M/(4\pi a^2)$. Put the particle at P with $OP = b < a$. If $b = 0$, every direction from O is the same and the pulls of opposite points of the shell cancel, so the force is zero. From now on take $0 < b < a$ and use the line OP as an axis, measuring distance along it from O in the direction of P.



Slice the shell into thin rings by planes perpendicular to OP. The ring between polar angles θ and $\theta+\delta\theta$ (measured at O from the direction OP) has radius $a\sin\theta$, centre at

position $a\cos\theta$ on the axis, and width $a\,\delta\theta$. Its mass is

$$\delta M = \frac{M}{4\pi a^2}\cdot 2\pi a\sin\theta\cdot a\,\delta\theta = \tfrac{1}{2}M\sin\theta\,\delta\theta.$$

The centre of this ring is at signed distance $a\cos\theta - b$ from P. By the ring result its pull on P is along the axis, and its component in the positive direction (away from O, through P) is

$$\frac{G\,\delta M\,(a\cos\theta - b)}{\left((a\cos\theta - b)^2 + a^2\sin^2\theta\right)^{3/2}} = \frac{G\,\delta M\,(a\cos\theta - b)}{(a^2 + b^2 - 2ab\cos\theta)^{3/2}}.$$

This sign convention is right in both cases: when $a\cos\theta > b$ the ring is beyond P and pulls in the positive direction; when $a\cos\theta < b$ it pulls the other way. Every ring pulls along the same axis, so the total force is along OP, with component

$$F = \frac{GM}{2}\int_0^{\pi}\frac{(a\cos\theta - b)\sin\theta}{(a^2 + b^2 - 2ab\cos\theta)^{3/2}}\,\mathrm{d}\theta.$$

? **Ask yourself** What substitution tames the denominator?

The quantity $a^2 + b^2 - 2ab\cos\theta$ is, by the cosine rule, the square of the distance s from P to the ring (the line marked s in the figure). So we substitute

$$s^2 = a^2 + b^2 - 2ab\cos\theta, \qquad 2s\,\mathrm{d}s = 2ab\sin\theta\,\mathrm{d}\theta.$$

As θ runs from 0 to π, s increases from $a - b$ to $a + b$ (positive because $b < a$). Also

$$a\cos\theta - b = \frac{a^2 + b^2 - s^2}{2b} - b = \frac{a^2 - b^2 - s^2}{2b}.$$

Using $\sin\theta\,\mathrm{d}\theta = \dfrac{s\,\mathrm{d}s}{ab}$,

$$\begin{aligned} F &= \frac{GM}{2}\int_{a-b}^{a+b}\frac{a^2 - b^2 - s^2}{2b}\cdot\frac{1}{s^3}\cdot\frac{s}{ab}\,\mathrm{d}s = \frac{GM}{4ab^2}\int_{a-b}^{a+b}\left(\frac{a^2 - b^2}{s^2} - 1\right)\mathrm{d}s \\ &= \frac{GM}{4ab^2}\left[-\frac{a^2 - b^2}{s} - s\right]_{a-b}^{a+b}. \end{aligned}$$

At $s = a + b$ the bracket is $-(a - b) - (a + b) = -2a$, and at $s = a - b$ it is $-(a + b) - (a - b) = -2a$. These are equal, so

$$F = \frac{GM}{4ab^2}\big(-2a - (-2a)\big) = 0.$$

The force is along OP and its component along OP is zero, so the particle experiences no gravitational force due to the shell, as required.

Checks

- Ring: at $x = 0$ the force is zero (the particle is at the centre and the pulls balance), and for $x \gg r$ it tends to Gm/x^2, as if all the mass were at O.
- Shell, outside: if instead $b > a$, the lower limit becomes $s = b - a$, where the bracket is $(a + b) - (b - a) = 2a$. Then $F = \frac{GM}{4ab^2}(-4a) = -GM/b^2$: an attraction towards O as if the whole mass were at the centre. That is Newton's other shell theorem, a good sign the integral is right.
- Numerically, summing 2000 point masses round a ring reproduced the formula to ten decimal places. Simpson's rule on the θ-integral gave about 10^{-15} for several random $b < a$, and a Monte Carlo shell of 2×10^5 random points with $a = 2$, $b = 1.3$, $GM = 1$ gave a force below 10^{-3}, against a typical scale $GM/a^2 = 0.25$.

Common slips

- Taking the ring's mass element as $M\,\delta\theta/\pi$. The rings near $\theta = \pi/2$ are bigger, so the mass is proportional to $\sin\theta\,\delta\theta$, not $\delta\theta$.
- Using $|a\cos\theta - b|$ in the ring formula. The rings on the two sides of P pull in opposite directions, and that cancellation is the whole point.
- Getting the limits of s wrong: at $\theta = 0$ the distance is $a - b$, not $b - a$, because P is inside.

Beyond STEP

This is Newton's shell theorem, and it has a slicker proof using Gauss's law for gravity: the flux of the gravitational field through any closed surface is proportional to the mass inside, so by symmetry the field inside an empty spherical shell must vanish. The same mathematics gives the potential, and the fact that the potential is constant inside the shell is an early example of a harmonic function, studied in first-year vector calculus. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats the shell theorem and gravitational potential carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.